

CODE 5

```
interface SmartDevice {  
    void turnOn();  
    void turnOff();  
}
```

```
class SmartFan implements SmartDevice {  
    @Override  
    public void turnOn() {  
        System.out.println("SMART FAN: TURNED ON");  
    }  
    @Override  
    public void turnOff() {  
        System.out.println("SMART FAN: TURNED OFF");  
    }  
}
```

```
class SmartLight implements SmartDevice {  
    @Override  
    public void turnOn() {  
        System.out.println("SMART LIGHT: TURNED ON");  
    }  
    @Override  
    public void turnOff() {  
        System.out.println("SMART LIGHT: TURNED OFF");  
    }  
}
```

```
class SmartAC implements SmartDevice {  
    @Override
```

```
public void turnOn() {  
    System.out.println("SMART AC: TURNED ON");  
}  
  
@Override  
public void turnOff() {  
    System.out.println("SMART AC: TURNED OFF");  
}  
}
```

```
class Main {  
    public static void main(String[] args) {  
        SmartDevice device;  
  
        device = new SmartFan();  
        device.turnOn();  
        device = new SmartFan();  
        device.turnOff();  
        device = new SmartLight();  
        device.turnOn();  
        device = new SmartLight();  
        device.turnOff();  
        device = new SmartAC();  
        device.turnOn();  
        device = new SmartAC();  
        device.turnOff();  
  
        device = new SmartFan();  
        device.turnOn();  
        device.turnOff();  
        device = new SmartLight();  
        device.turnOn();  
    }  
}
```

```
device.turnOff();  
device = new SmartAC();  
device.turnOn();  
device.turnOff();
```

```
device = new SmartFan();  
device.turnOn();  
device.turnOff();  
device = new SmartLight();  
device.turnOn();  
device.turnOff();  
device = new SmartAC();  
device.turnOn();  
device.turnOff();
```

```
device = null;  
try {  
    device.turnOn();  
}  
catch (NullPointerException e) {  
    System.out.println("NULL INTERFACE REFERENCE");  
}
```

```
System.out.println("SmartFan must implement all interface methods");  
System.out.println("Interface SmartDevice cannot be instantiated");  
System.out.println("NULL interface reference causes NullPointerException");  
System.out.println("turnON() is incorrect. Correct method is turnOn()");  
System.out.println("Method signature must be turnOn()");  
System.out.println("SmartFan must use implements SmartDevice");  
System.out.println("Interface methods must be implemented as public");  
}
```

}

OUTPUT:

```
SMART FAN: TURNED ON
SMART FAN: TURNED OFF
SMART LIGHT: TURNED ON
SMART LIGHT: TURNED OFF
SMART AC: TURNED ON
SMART AC: TURNED OFF
SMART FAN: TURNED ON
SMART FAN: TURNED OFF
SMART LIGHT: TURNED ON
SMART LIGHT: TURNED OFF
SMART AC: TURNED ON
SMART AC: TURNED OFF
SMART FAN: TURNED ON
SMART FAN: TURNED OFF
SMART LIGHT: TURNED ON
SMART LIGHT: TURNED OFF
SMART AC: TURNED ON
SMART AC: TURNED OFF
NULL INTERFACE REFERENCE
SmartFan must implement all interface methods
Interface SmartDevice cannot be instantiated
NULL interface reference causes NullPointerException
turnON() is incorrect. Correct method is turnOn()
Method signature must be turnOn()
SmartFan must use implements SmartDevice
Interface methods must be implemented as public

...Program finished with exit code 0
Press ENTER to exit console.█
```